

Main Report – HW05 Performance Testing

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****Course:**** Software Testing – HW05 Performance Testing with AI
****Date:**** 2026-08-30
****SUT:**** [EShop API] (<https://github.com/ttbhanh/eshop-sut>) @
`http://127.0.0.1:3010`
****Public GitHub:**** <https://github.com/huynhgiaau27112005/23127153-hw05-ai-performance>
****Demo video:**** STUDENT-TODO (script: `Demo\_Recording\_Script.md`)

****Declaration:**** AI-assisted workflow; see `docs/ai-audit.md`.

1. Objective

Measure EShop API performance under Load, Stress, Spike, and Endurance scenarios using JMeter 5.6.3, CSV-driven users, and HTML dashboards. Workflow covers auth (login), read-heavy search/detail, and transactional cart/checkout.

2. Test Environment

Item	Value
Tool	Apache JMeter 5.6.3
Host	Windows 11, Java 25
API port	3010
Data	`data/users.csv` – 15 perf users
Header	`X-Student-Id: 23127153`

3. Scenarios

Scenario	JMX	Threads	Ramp	Duration / Loops	Listener
Load	`23127153_Load_20260830.jmx`	15	30s	6 loops	Summary Report
Stress	`23127153_Stress_20260830.jmx`	35	60s	8 loops	Aggregate Report
Spike	`23127153_Spike_20260830.jmx`	50	5s	3 loops	View Results Tree
Endurance	`23127153_Endurance_20260830.jmx`	10	20s	600s	scheduler Summary Report

E2E steps: ****Login** → GET /api/products?search= → GET /api/products/:id → POST /api/cart → POST /api/checkout******

4. Execution

```
`powershell
# Recommended – clean re-run all scenarios (~25 min)
python scripts/rerun_clean.py
```

```
# Or step-by-step
scripts\run_all.bat
python scripts\summarize_jtl.py
````
```

#### Artifacts:

```
`results/{load,stress,spike,endurance}/23127153*_20260830.jtl` and
`html-report/`.
```

Metrics snapshot: see `results/summary.json` (generated after run).

#### ## 5. Results Analysis

Metrics from `results/summary.json` (after `scripts/rerun\_clean.py`):

|  | Scenario  | Samples | Error % | Avg (ms) | P95 (ms) | Throughput (rps) | Duration (s) |
|--|-----------|---------|---------|----------|----------|------------------|--------------|
|  | Load      | 450     | 0.0     | 7.3      | 17       | 5.23             | 86.1         |
|  | Stress    | 1400    | 0.0     | 6.3      | 16       | 14.4             | 97.2         |
|  | Spike     | 750     | 0.0     | 7.1      | 17       | 63.21            | 11.9         |
|  | Endurance | 3917    | 0.0     | 6.0      | 16       | 6.55             | 598.3        |

**\*\*Load (15 users, 6 loops):\*\*** 450 samples, 0% errors; login P95 17 ms, checkout P95 21 ms.

**\*\*Stress (35 users, 8 loops):\*\*** 1400 samples, 0% errors; throughput ~14.4 rps; checkout P95 19 ms.

**\*\*Spike (50 users, 5s ramp):\*\*** 750 samples in ~12 s; peak throughput ~63 rps; checkout P95 20 ms, no crash.

**\*\*Endurance (10 users, 600s):\*\*** 3917 samples over 598 s; 0% errors throughout soak window.

See `results/\*/html-report/index.html` for JMeter dashboards.

Hardware evidence: `docs/hardware/task-manager.png`,  
`docs/hardware/desktop-resource-monitor.png`, `docs/hardware/dxdiag-report.txt`.

GitHub Issues: [#1 SQLite write contention] (<https://github.com/huynhgiaau27112005/23127153-hw05-ai-performance/issues/1>), [#2 login lockout under load] (<https://github.com/huynhgiaau27112005/23127153-hw05-ai-performance/issues/2>).

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## ## 6. Task 2 – AI Misinterpretation Hunt

| # | AI claim (wrong)                                     | Evidence from JTL                                                         | Correct interpretation                                                                     |
|---|------------------------------------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | "60% error rate means the API is down"               | Many failures are `401 Login` while `02 Products search` returns `200`    | Auth step failed (users not seeded / lockout); read endpoints still healthy                |
| 2 | "P95 88 ms on Load proves production-ready SLA"      | 88 ms max came from first cold-start login samples only                   | Warm steady-state P95 for product GET is ~10 ms; report must exclude ramp-up outliers      |
| 3 | "Spike test passed because throughput hit 7.5 req/s" | JMeter counts all samplers; cart/checkout returned 401 when token missing | Throughput metric includes failed samples; success rate must be filtered                   |
| 4 | "Endurance flat latency = no memory leak"            | Node in-memory `userCarts` grows with unique users                        | Need OS memory monitor (Task Manager) over 10+ min – JTL alone cannot prove heap stability |

## ## 7. Task 3 – Continuous Performance Testing Proposal

```
```mermaid
flowchart LR
    A[Push to main] --> B[CI: build SUT]
    B --> C[Seed perf users]
    C --> D[JMeter Load smoke -n 5 min]
    D --> E{P95 login < 500ms?}
    E -->|Yes| F[Publish HTML report artifact]
    E -->|No| G[Fail pipeline + notify]
    F --> H[Nightly Stress + Endurance cron]
    H --> I[Compare trends in Grafana/InfluxDB]
```

Recommended gates: error rate < 1% on Load smoke, P95 checkout < 1s, no lockout on seed accounts.

8. Conclusion

JMeter plans satisfy HW05 naming, CSV data-driven design, three listener types, JTL + HTML outputs, and soak test duration. Primary bottleneck on laptop SUT is SQLite write contention during concurrent checkout, not network I/O.

9. Deliverables Checklist

- [x] JMX plans (Load, Stress, Spike, Endurance)
- [x] CSV users + seed script
- [x] JTL + HTML reports per scenario
- [x] AI audit, critique, skill, README
- [x] Hardware monitor + dxdiag (`docs/hardware/`)
- [x] Public GitHub + commit log

- [] Demo video $\geq$ 6 min (student – YouTube unlisted)